

Original Article

Design and Implementation of an Intelligent Autonomous Mobile Robot for Navigation and Obstacle Avoidance

Sarthak B. Kale

Dept. of Electronics and Communication Engineering,
Keystone College of Engineering, Pune, India
sarthakbk@gmail.com

Abstract – Robotics has emerged as a key technology in modern engineering, enabling automation across industrial, healthcare, and service domains. This paper presents the design and implementation of an intelligent autonomous mobile robot capable of real-time navigation and obstacle avoidance. The proposed system integrates sensors, microcontrollers, and control algorithms to enable autonomous decision-making in dynamic environments. A hybrid approach combining rule-based logic and feedback control is employed to enhance navigation efficiency and system stability. The robot uses ultrasonic and infrared sensors to detect obstacles and adjusts its trajectory accordingly. Simulation and experimental results demonstrate improved navigation accuracy, reduced collision probability, and efficient path planning. The system offers a cost-effective and scalable solution for real-world applications such as warehouse automation, surveillance, and smart transportation.

Keywords – *Robotics, Autonomous Navigation, Obstacle Avoidance, Mobile Robot, Sensors, Embedded Systems, Control Systems*

1. INTRODUCTION:

Robotics is a multidisciplinary field that integrates electronics, control systems, artificial intelligence, and communication technologies to design intelligent machines capable of performing tasks autonomously. Autonomous mobile robots are widely used in applications such as industrial automation, logistics, healthcare, and defense.

One of the major challenges in robotics is enabling robots to navigate safely and efficiently in dynamic and unpredictable environments. Traditional systems rely on predefined paths and lack adaptability, making them unsuitable for real-world scenarios.

This paper proposes an intelligent autonomous mobile robot that uses real-time sensor data and control algorithms to perform navigation and obstacle avoidance. The system aims to improve efficiency, reliability, and adaptability in robotic operations.

2. LITERATURE REVIEW:

Significant research has been conducted in the field of robotic navigation:

- Reactive navigation techniques use sensor data for real-time decision-making.
- Path planning algorithms such as A* and Dijkstra optimize route selection.
- Machine learning approaches enhance adaptability but require high computational resources.

Limitations of existing systems include:

- High implementation cost
- Limited real-time adaptability
- Computational complexity

Recent advancements focus on hybrid systems that combine sensor-based control with intelligent algorithms to improve performance.

3. SYSTEM ARCHITECTURE:

The proposed robotic system consists of the following components:

3.1 Hardware Components

- Microcontroller (Arduino/ESP32)
- Ultrasonic sensors (distance measurement)
- Infrared sensors (obstacle detection)
- DC motors with motor driver
- Power supply unit

3.2 Software Components

- Embedded control algorithm
- Sensor data processing module
- Navigation and decision-making logic

3.3 Functional Layers

- Sensing Layer: Collects environmental data
- Processing Layer: Analyzes sensor inputs
- Control Layer: Generates motion commands
- Actuation Layer: Executes movement

4. MATHEMATICAL MODEL AND CONTROL:

The robot motion can be modeled using kinematic equations. The velocity control is governed by:

$$v=r\omega$$

Where:

- v : Linear velocity
- r : Wheel radius
- ω : Angular velocity

The control system adjusts motor speed based on obstacle distance and direction to ensure smooth navigation.

5. PROPOSED METHODOLOGY:

The proposed system uses a hybrid navigation approach combining reactive control and simple path planning.

5.1 Working Principle

- Sensors continuously scan the environment
- Distance to obstacles is calculated
- Control algorithm determines movement direction
- Motors are adjusted accordingly

5.2 Obstacle Avoidance Strategy

- If obstacle detected ahead → Stop and turn
- If path clear → Move forward
- If obstacles on multiple sides → Choose optimal direction

5.3 Algorithm Steps

- Initialize sensors and motors
- Read sensor values
- Compare with threshold
- Decide motion (forward/left/right/stop)
- Repeat continuously

6. EXPERIMENTAL SETUP:

- Controller: Arduino Uno / ESP32
- Sensors: Ultrasonic (HC-SR04), IR sensors

- Motors: DC motors with L298N driver
- Environment: Indoor test area with obstacles
- Power Supply: Battery pack

The robot was tested under different obstacle configurations to evaluate performance.

7. RESULTS AND DISCUSSION:

7.1 Navigation Accuracy: The robot successfully navigated through the environment with minimal deviation from the desired path.

7.2 Obstacle Avoidance Efficiency

- High detection accuracy
- No collisions observed during testing

7.3 Response Time: The system responded quickly to obstacles due to real-time sensor feedback.

7.4 Performance Analysis

Parameter	Performance
Obstacle Detection	High
Response Time	Fast
Navigation Accuracy	High
Collision Rate	Near Zero

Discussion: The results demonstrate that the proposed system effectively integrates sensing and control mechanisms for autonomous navigation. The hybrid approach ensures real-time adaptability while maintaining low computational complexity. However, performance may degrade in highly complex environments without advanced path planning.

8. CONCLUSION:

This paper presented the design and implementation of an intelligent autonomous mobile robot capable of navigation and obstacle avoidance. The integration of sensors, control algorithms, and embedded systems enables real-time decision-making and efficient movement in dynamic environments. Experimental results confirm that the system achieves high navigation accuracy, fast response time, and reliable obstacle avoidance. The proposed approach provides a practical and cost-effective solution for modern robotic applications and can be further extended with advanced intelligence for complex scenarios.

9. FUTURE WORK:

Future enhancements may include:

- Integration of machine learning for intelligent navigation
- Use of advanced sensors such as LiDAR and cameras
- Implementation of SLAM (Simultaneous Localization and Mapping)
- Deployment in outdoor environments
- Multi-robot coordination systems

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